

Question

Hydrated inorganic salt **X** (chemical formula $\text{MY}_x \cdot y\text{H}_2\text{O}$) can be obtained by dissolving the element **M** in an aqueous solution of common inorganic acid **HY** and crystallizing, where the mass fraction of **M** is 26.30%. When **X** is heated, anhydrous $\text{MY}_x \cdot y\text{H}_2\text{O}$ cannot be obtained. Instead, it loses 50.31% of its weight at 80 °C and transforms into **A**. **A** loses another 16.77% of its weight at 180 °C (relative to **X**) to obtain black oxide **B**. It is known that the oxidation state of **M** does not change during the entire weight loss process, and the mixture of gases released during both weight loss events contains water vapor and a colored, irritating gas. Let a be the sum of the stoichiometric coefficients on both sides of the equation for the first weight loss, and b be the sum of the stoichiometric coefficients on both sides of the equation for the second weight loss, then a/b is

A. All other options are incorrect

B. 1.33

C. 1.47

D. 2.31

E. 4.80

F. 0.94

G. 1.88

H. 2.93

I. 0.87

J. 0.65

K. 0.42

L. 1.28

M. 2.50

N. 3.04

O. 3.33

P. 4.23

Q. 3.77

R. 3.53

S. 1.96

Answer

Correct Answer: **S**

Detailed Explanation

From the mass fraction of **M** and the information on weight loss, it can be seen that the mass fraction of **M** in oxide **B** is $\frac{0.2630}{1-0.5031-0.1677} = 0.7989$, so the mass fraction of oxygen in **B** is 20.11%.

If we assume the chemical formula of **B** is MO_a , then the atomic weight of M is $16a \times 0.7989 / 0.2011$

CHECKPOINT

1 PTS

Assume the chemical formula of **B** is MO_a , then the atomic weight of M is $16a \times 0.7989 / 0.2011$

Enumeration shows that when $a = 1$, $M_M = 63.56 \text{ g} \cdot \text{mol}^{-1}$, corresponding to Cu

CHECKPOINT

1 PTS

M is Cu

Substituting the mass fraction of **M** in **X**, we know that the molecular weight of **X** is 241.63, and the remaining molecular weight after deducting **M** is about 178.1. Since **HY** is a monobasic acid, and considering the common oxidation state of copper, it can be inferred that $x = 2$; at this time, since **M** can be dissolved in **HY** to obtain **X**, and a colored irritating gas is released during weight loss, it can be boldly guessed that **HY** is nitric acid. Substituting the remaining molecular weight and deducting the formula weight of two nitrate radicals, it can be known that the number of water molecules is exactly 3, which also verifies the guess that **HY** is nitric acid. Therefore, **X** is $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$

CHECKPOINT

2 PTS

X is $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$

Since the oxidation state of Cu does not change during the weight loss process, the decomposition of **X** can be regarded as the hydrolysis of copper ions, and heating causes **X** to lose nitric acid molecules and water molecules, while **A** is a basic copper nitrate.

From the mass fraction of weight loss, the molecular weight of the first weight loss is $241.63 \times 0.5031 = 121.56$, and the second is $241.63 \times 0.1677 = 40.52$ (if you are sensitive enough to the data, you will find that the ratio between the two is very close to 3:1).

The molecular weight of nitric acid is 63.018, and that of water molecule is 18.016. There is no solution in the case of 1 Cu, so consider expanding the multiple to 2 Cu, weight loss 243.12, corresponding to $3\text{HNO}_3 + 3\text{H}_2\text{O}$; and the second is $\text{HNO}_3 + \text{H}_2\text{O}$. Therefore, **A** is $\text{Cu}_2(\text{OH})_3\text{NO}_3$

CHECKPOINT

2 PTS

A is $\text{Cu}_2(\text{OH})_3\text{NO}_3$

First weight loss: $8[\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}] = 4\text{Cu}_2(\text{OH})_3\text{NO}_3 + 3\text{O}_2 + 12\text{NO}_2 + 18\text{H}_2\text{O}$

CHECKPOINT

1.5 PTS

First weight loss: $8[\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}] = 4\text{Cu}_2(\text{OH})_3\text{NO}_3 + 3\text{O}_2 + 12\text{NO}_2 + 18\text{H}_2\text{O}$

Second weight loss: $4\text{Cu}_2(\text{OH})_3\text{NO}_3 = 8\text{CuO} + \text{O}_2 + 4\text{NO}_2 + 6\text{H}_2\text{O}$

CHECKPOINT

1.5 PTS

Second weight loss: $4\text{Cu}_2(\text{OH})_3\text{NO}_3 = 8\text{CuO} + \text{O}_2 + 4\text{NO}_2 + 6\text{H}_2\text{O}$

$a = 45$, $b = 23$, $a/b = 1.96$, select S

CHECKPOINT

1 PTS

$a/b = 1.96$